

LogiFlow — Weekly Progress Report

Week 4 | NextGen Forge Technologies Internship | Ref: NFGT/HR/INT/2026/160

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Project:	LogiFlow — Enterprise Logistics & Warehouse Management Platform		

1. Tasks Completed This Week

Task W4.1 — Vehicle, Driver & Dispatch Planning APIs

- Implemented full Vehicle Management CRUD (create, list with status/type filters, list available vehicles, update, soft-delete to Maintenance status).
- Implemented full Driver Management CRUD (create, list with availability filter, availability toggle, soft-delete).
- Implemented Dispatch Planning APIs (create, list with date/status filters, detail, complete) with availability validation preventing assignment of an In-Use vehicle or unavailable driver.
- Verified state transitions: Dispatch creation flips Vehicle to In-Use and Driver to unavailable, and links Shipment to Assigned; Dispatch completion reverses all three and marks Shipment as Delivered.

Task W4.2 — Analytics, Activity Logs, Notifications & Support Ticket APIs

- Built the analytics summary endpoint covering total active orders, shipments grouped by status, warehouse capacity utilisation, top 5 products by order volume, and dispatches completed in the last 7 days.
- Built the orders-by-day endpoint returning a 30-day order count series for the dashboard line chart.
- Exposed the Activity Log viewer endpoint (Administrator-only) with filtering by entity, user, and date range; verified existing log entries from Inventory, Order, and Shipment actions display correctly.
- Implemented Notification endpoints: fetch current user's unread notifications, mark single notification as read, and mark all as read.
- Implemented full Support Ticket module: create ticket, list tickets (self-filtered to assigned tickets for non-Administrator roles), assign ticket (Administrator-only), update ticket status, and append comments to the ticket timeline.

Task W4.3 — Week 4 Backend Integration Review

- Updated README.md with complete API reference documentation covering all 14 route groups, role restrictions, status enum reference, and system/business-flow diagrams.
- Committed all Week 4 work to the develop branch and opened a pull request into main.

2. Testing Performed

- Postman testing of all Vehicle, Driver, and Dispatch endpoints, including negative tests for reassigning an already-In-Use vehicle and an unavailable driver, and invalid/nonexistent shipment, vehicle, and driver IDs.

- End-to-end dispatch lifecycle verified: shipment creation through dispatch assignment, completion, and confirmation that vehicle/driver availability and shipment status update correctly at each step.
- Verified Analytics summary and orders-by-day endpoints return accurate figures against known seeded test data.
- Verified Activity Log filtering by entity, user, and date range, and pagination correctness.
- Verified Notification read/unread flow and mark-all-as-read behaviour.
- Verified full Support Ticket lifecycle: creation, assignment, status update, and comment threading.

3. Blockers Faced

- Recurring pattern of missing await on Mongoose query calls and copy-paste variable/model name mismatches across new controllers — caught and resolved during self-review and testing.
- Field name mismatches between schema definitions and controller code (e.g. Driver licence number spelling) caused a duplicate-key error on a unique index; resolved by aligning naming and clearing the affected collection.
- Ticket model did not originally include an assigneeId field required for the assignment endpoint — added retroactively before implementing the assign controller.

4. Known Gaps / Notes

- Activity Log writes currently cover Inventory, Order, and Shipment actions only; Vehicle, Driver, and Dispatch actions are not yet audit-logged.
- Order and Ticket status transitions are not currently enforced as strict forward-only state machines at the API level (unlike Shipment status).
- Full 5-role permission-matrix testing was scoped down to representative write-endpoint sampling rather than exhaustive per-module testing across all five roles.

5. Plan for Next Week

- Begin Week 5: initialize the React (Vite) frontend, configure Tailwind CSS, set up the Axios instance with JWT interceptor.
- Build authentication UI (Login screen), AuthContext, and ProtectedRoute wrapper.
- Build the application shell — sidebar, top navigation, role-conditional menu rendering.
- Begin the KPI dashboard and analytics chart integration using the Week 4 analytics endpoints.

Submitted by Aryan Ajmera to hr@nextgenforgetechnologies.com